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A quantitative analysis of ATF surface characteristics on critical heat flux using Machine learning

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ABSTRACT

The effects of surface characteristics on pool boiling Critical Heat Flux (CHF) are qualitatively understood based on previous investigations. However, more quantitative analyses are needed since the existing CHF correlations do not provide good predictions for modified surfaces. Using machine learning (ML) models as a tool, this study performed a quantitative analysis of relevant CHF parameters under pool boiling conditions: pressure, a dimensional feature, average roughness, static contact angle, surface orientation, and substrate thermal effusivity. A database was constructed by collecting accident tolerant fuels (ATF) CHF experimental data available from fourteen published studies. After hyperparameter optimization, the random forest (RF) model was selected for achieving the best fitting scores relative to other tested models. Feature importance models ranked static contact angle and pressure as the most important features, which is consistent with some of the literature CHF predictive models that take the surface characteristics into consideration. Finally, CHF predictions were obtained and compared to CHF experimental data. The effect of each feature on CHF was analyzed, while keeping other features fixed, by observing the experimental and predicted datapoints trends. The RF model demonstrated the ability to capture the experimental data trends, showing the RF model is suitable as a predictive pool boiling CHF model for this database.

1. Introduction

Accident-tolerant fuels (ATFs) have been widely studied in the past decade in order to improve the thermal–mechanical and oxidation performance of the current cladding materials used in light water nuclear reactors (LWRs). One of the main goals of the ATF campaign is to develop and test materials which are more resilient to severe accident conditions. In pursuit of this goal, several materials have been proposed to modify (*e.g.*, Cr-coated) or replace (*e.g.*, SiC and FeCrAl) the most widely used nuclear cladding material, Zircaloy, whose reaction with water at high temperatures produces hydrogen gas, which is highly inflammable at low concentrations. Even though the primary focus of previous studies has been the material oxidation reaction, improvements in heat transfer efficiency are sought after, as these would allow for larger safety margins during reactor operation.

In the 1950 s and 1960 s, authors were already aware that surface characteristics had some effect on pool boiling heat transfer (Corty and Foust, 1955; Griffith, 1958; Kurihara and Myers, 1960). These early studies served as a foundation for more recent investigations where the effect of different surface characteristics, such as geometry (Tachibana et al., 1967; Kam et al., 2015; Seo et al., 2015; Yeom et al., 2020; He et al., 2022; He and Chen, 2023; Jo et al., 2019), roughness (Yeom et al., 2020; He et al., 2022; Fong et al., 2001; Son et al., 2015; Son et al., 2019; Ali et al., 2020), contact angle (Kam et al., 2015; Seo et al., 2015; Yeom et al., 2020; He et al., 2022; He and Chen, 2023; Jo et al., 2019; Son et al., 2015; Son et al., 2019; Ali et al., 2020; Fong et al., 2003; Ahn et al., 2010; Lee et al., 2014; Ali et al., 2018; Kim et al., 2019; Cheol Lee et al., 2019; Kam et al., 2021; Zhang et al., 2023), capillary wicking (surface tension forces) (Son et al., 2019; Ahn et al., 2010; Kim et al., 2019; Zhang et al., 2023; Ahn et al., 2012; Son et al., 2017; Son et al., 2018; Son et al., 2020), heating element thermal properties (Tachibana et al., 1967; Yeom et al., 2020; He et al., 2022; Jo et al., 2019), and different coating techniques (Tachibana et al., 1967; Kam et al., 2015; Yeom et al., 2020; He et al., 2022; He and Chen, 2023; Son et al., 2015; Son

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et al., 2019; Ali et al., 2020; Fong et al., 2003; Ahn et al., 2010; Lee et al., 2014; Ali et al., 2018; Kim et al., 2019; Cheol Lee et al., 2019; Kam et al., 2021; Zhang et al., 2023; Ahn et al., 2012; Son et al., 2018; Son et al., 2020), were individually investigated.

The effects of roughness and contact angle on critical heat flux (CHF) are well described by Kim *et al.* (Kim et al., 2018). This study shows that for hydrophobic surfaces, CHF decreases exponentially, at a rate related to the cosine of the contact angle, while CHF stays consistent with Kandlikar's prediction (Kandlikar, 2001) for hydrophilic surfaces. It is important to highlight that roughness and contact angle are related surface characteristics parameters (Wenzel, 1936). Contact angle, a parameter that reflects the wettability of a surface, affects CHF inversely, meaning CHF increases when contact angle decreases, and vice versa. However, when the surface becomes extremely hydrophilic, its contact angle approaches zero and this parameter becomes effectively useless. In these cases, the wicking effect becomes a more relevant parameter, and alternative methods to measure the liquid spreading on the surface such as liquid wet column imaging (Do Nascimento et al., 2019) and dry area spreading imaging (Son et al., 2018) are used. Rewetting speed is directly related to the capillary wicking. When capillary forces increase, fluid spreads much quicker across the heating element. This postpones all the heat transfer regimes, including the CHF phenomena, due to the faster dry spot rewetting after vapor bubble departure from the heating element cavities.

Arik and Bar-Cohen (Arik and Bar-Cohen, 2003) analyzed the effect of the thermal activity parameter (S), which is obtained by multiplying the thermal effusivity of the heating element by its thickness, and observed CHF improvement achieves an asymptotic maximum value. Later, Jo *et al.* (Jo et al., 2019) discussed two non-hydrodynamic parameters that combine the heating element's thermal properties with its geometry (thickness), coming to an understanding that thermal properties do not significantly affect CHF. In another study (Ham et al., 2017), it was observed that a thick porous layer, created by nanoparticle deposition, affected CHF negatively due to both an increase in thermal resistance and a decrease in the number of cavities.

Surface modifications, such as pre-oxidation and coatings have been explored in CHF studies using various techniques, such as direct current (DC) magnetron sputtering (Son et al., 2015; Son et al., 2019; Kim et al., 2019), anodization (Ahn et al., 2010; Kam et al., 2021), high-temperature dry air oxidation (Tachibana et al., 1967; Fong et al., 2003; Lee et al., 2014; Kim et al., 2019; Cheol Lee et al., 2019), electroplating (Kam et al., 2015), physical vapor deposition (PVD) sputtering (Kam et al., 2015), cold spray technique (Yeom et al., 2020; Jo et al., 2019), autoclave test (under a harsh environment) (Ali et al., 2018); *etc*. These techniques involve different approaches and even the same technique can be applied differently, varying the time, temperature, heat flux, *etc*. Therefore, it is difficult to compare data provided by studies focused on different surface modification techniques.

Since the available correlations and discussions do not reach a consensus on which parameters are the most important affecting pool boiling heat transfer, this study aims to use different machine learning (ML) models to, first, quantify and rank the parameters' effects on pool boiling CHF, and second, obtain and compare CHF predictions from the best ML model trained with experimental pool boiling CHF values.

2. Database construction

In this section, the database and the pre-analysis steps used to train and test the ML models are described. These steps are very important as the performance of ML models is highly dependent on the quality of the input data. Initially, ATF CHF data was collected from 19 studies available in the literature with a total of 221 datapoints. Then, the collected data was filtered based on an analysis of the number of available data for the input parameters selection. The decision of which input parameters (features) would be used to train and test the ML models was highly dependent on the data available in the studies. Different studies focus on how different parameters affect CHF. Based on the available data, this study included the largest common set of inputs across different datapoints: pressure, dimensional feature, average roughness, static contact angle, surface orientation, and heating element thermal effusivity. Fig. 2.1 shows the number of CHF datapoints for each measured parameter.

After data filtering, a total of 101 datapoints from 14 different studies was used, for water-based pool boiling experiments at saturated atmospheric conditions only. Since most papers do not mention the experimental temperature and pressure, the main author's location was primarily considered, and its elevation was used to determine the local pressure and the related saturation temperature of the experiments. To minimize any erroneous considerations, this approach was carefully evaluated on a case-by-case basis. The two parameters (*i.e.*, experiment location and elevation) are important since they affect all the fluid properties, which are known to have an important contribution on boiling heat transfer. Most of the experiments were performed close to sea level, but a few of them were performed at relatively high elevations. Table 2.1 presents the studies with the considered experiment's location, its related elevation, calculated atmospheric pressure, and saturation temperature.

As seen in Fig. 2.1, while pressure, surface orientation, and the dimensional feature (based on the heating element geometry) are provided in all studies, receding contact angle and peak valley distance are only provided in a small number of studies. Even though they are important parameters for pool boiling heat transfer, they cannot be used as inputs for the ML models due to the limited amount of data. For example, the receding contact angle is known as an important parameter in pool boiling heat transfer, as it is used in one of the most well-known CHF correlations, Kandlikar (Kandlikar, 2001), Eq. 1:

$$q''_{CHF} = \rho_g h_{fg}\left(\frac{1+cos\beta}{16}\right)\left[\frac{2}{\pi}+\frac{\pi}{4}(1+cos\beta)cos\phi\right]^{1/2}\left[\frac{\sigma g(\rho_f-\rho_g)}{\rho_g^2}\right]^{1/4}$$

where ϕ is the surface orientation angle, β is the receding contact angle, σ is the surface tension in N.m^{-1}, ρ_f and ρ_g are the saturated liquid and saturated vapor densities, respectively, in kg.m^{-3}, h_{fg} is the enthalpy of vaporization in J.kg^{-1}, and g is the acceleration due to the gravitational force in m.s^{-2}.

The dimensional feature is a normalized value given by dividing of one of the heating element's dimensions by capillary length. The heating element's dimension changes for different geometries: plates, discs, wires, vertical rods, horizontal rods, *etc.*, Table 2.2. The capillary length is a factor that balances the surface tension and the gravitational forces. It is solely related to the fluid properties, then it is affected by the different experiment pressures, as shown in Eq. 2.

$$L_c = \sqrt{\frac{\sigma}{(\rho_f-\rho_g)g}}$$

When determining the value for the heating element's thermal effusivity, the thermal effusivity of the most external layer of the heating element before CHF was considered. For example, if the sample was a bare sample, its material thermal properties were considered. If the sample was coated by any coating technique, the thermal properties of the coating material were considered. If instead, an oxide layer was the most external layer, the thermal properties of the oxide layer were considered. The thermal effusivity (ε_{th}) is calculated using the thermal properties of the material, in Eq. 3:

$$\varepsilon_{th} = \frac{k}{\sqrt{\alpha}} = \sqrt{k\rho c_p}$$

where k is the thermal conductivity in W.m^{-1}.K^{-1}, c_p is the specific heat capacity in J.kg^{-1}.K^{-1}, and α is the thermal diffusivity in m^2.s^{-1}. Thermal effusivity reflects how a material transfers energy to the

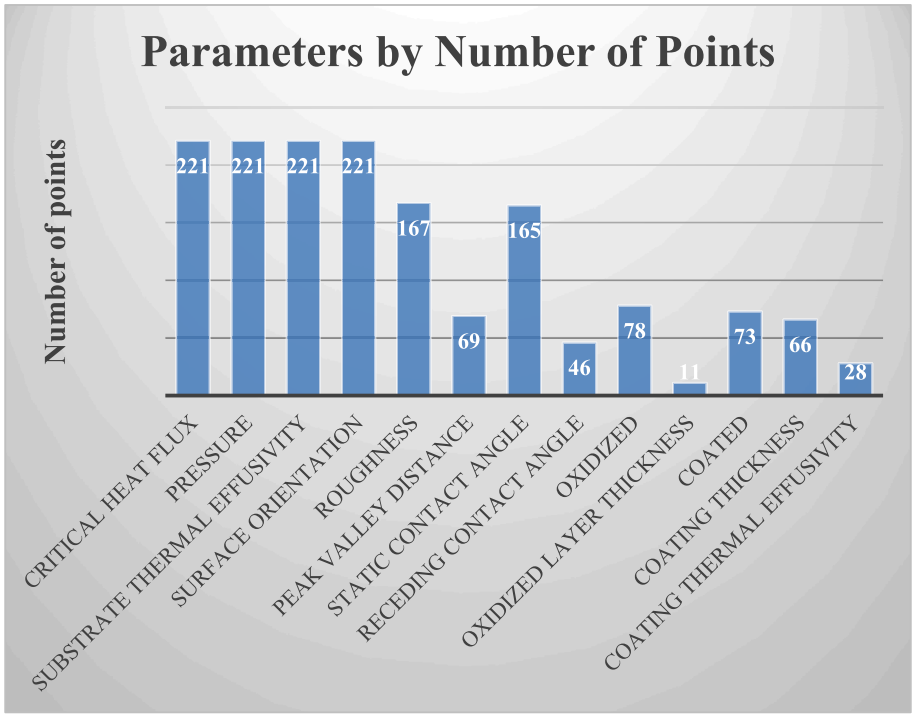


Fig. 2.1. Initial analysis of the potential input parameters by number of datapoints available.

Table 2.1
Literature studies used for database construction.

Location	Elevation [m]	Pressure [MPa]	Saturation Temperature [°C]	Number of datapoints	Reference
UW-Madison – Madison, WI, USA	272	0.0981	98.99	4	(Jo et al., 2019)
UW-Madison – Madison, WI, USA	272	0.0981	98.99	9	(Yeom et al., 2020)
UNM – Albuquerque, NM, USA	1576	0.0838	94.63	26	(He et al., 2022)
MIT- Cambridge, MA, USA	8	0.1012	99.95	6	(Zhang et al., 2023)
Hanyang University, Seoul, Korea	46	0.1008	99.81	7	(Kim et al., 2019)
UNM – Albuquerque, NM, USA	1576	0.0838	94.63	4	(Ali et al., 2018)
Hanyang University, Seoul, Korea	46	0.1008	99.81	6	(Son et al., 2019)
Hanyang University, Seoul, Korea	46	0.1008	99.81	2	(Seo et al., 2015)
Korea Advanced Institute of Science and Technology, Daejeon, Korea	53	0.1007	99.78	5	(Kam et al., 2015)
UNM – Albuquerque, NM, USA	1576	0.0838	94.63	5	(He and Chen, 2023)
UNM – Albuquerque, NM, USA	1576	0.0838	94.63	6	(Ali et al., 2020)
POSTECH, Pohang, Korea	37	0.1009	99.84	4	(Cheol Lee et al., 2019)
Canadian Nuclear Laboratories, Chalk River, ON, Canada	136	0.0997	99.48	7	(Fong et al., 2003)
POSTECH, Pohang, Korea	37	0.1009	99.84	10	(Ahn et al., 2010)

surrounding medium during a transient and has units $W.s^{1/2}.m^{-2}.K^{-1}$. A summary of the parameters selected as the input parameters (features) for the ML models is presented in Table 2.3 with their respective ranges.

After careful inspection of the available data, the following set of parameters were eventually selected: pressure, a dimensional feature, average roughness, static contact angle, surface orientation, and heating element thermal effusivity.

3. Machine learning models

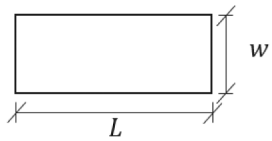
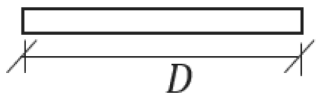
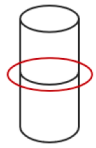
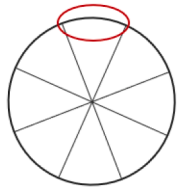
For this study, four different ML models were selected and constructed: (1) support vector regressor (SVR), (2) random forest (RF), (3) multi-layer perceptron (MLP), and (4) gaussian process (GP). These models were selected due to their powerful predictive performance and ability to capture both linear and nonlinear relationships.

SVR models are kernel-based models where a hyperplane and its related ε-insensitive tube are adjusted to the data. Datapoints are used as supporting vectors for the tube's limits and error is minimized according to the number of datapoints left outside the tube, called the slack variable, and their respective distances to the tube.

RF models are tree-based models where a set of decision trees is created and trained by implementing randomized processes, such as bootstrap sampling and feature selection. Because it is composed of a number of decision trees, the model is more generalized and less conditioned than any single decision tree.

MLP models are neural networks where the number of hidden layers, the layers in between the input and the output layers, is greater than one. The input of a neuron is a linear combination of the previous layer's outputs. Each neuron contains an activation function that transforms the linear combination output by implementing a specified mathematical function. After each set of iterations, known as a batch, the weights and biases present in the neural network are updated. The process is repeated for a specified number of epochs, when the whole dataset is used for training, or until a convergence criterion is achieved. The

Table 2.2
Dimensional feature selection based on heating element's geometry.

Geometry	Representation	Dimensional Feature
plate		width (w)
disc		diameter (D)
wire		πD
vertical rod		πD
horizontal rod		$\frac{\pi D}{8}$

authors' previous study (Serrao et al., 2023) provides a detailed description of these three models.

GP models are collections of random variables in which any finite number has a joint Gaussian distribution. Intuitively, this modeling framework can be interpreted as imposing a prior distribution over all possible function values. Then, based on the observed set of outputs, the initial beliefs are updated, obtaining a new posterior distribution. GPs models are known to be a powerful and flexible tool that can model complex trends through the specification of a covariance function (*i.e.*, kernel function). The kernel function is a key parameter that controls the generalization properties of the GP by defining the correlation between the datapoints. The kernel function is affected by three hyperparameters: the length scale parameter (ρ), the variance parameter of the kernel function (σ^2), and the stability parameter (α) that is added to the kernel matrix during model fitting. In this study, the length scale, the stability parameter, and the kernel function were optimized by trying out various popular kernel functions such as the Matern kernel and the radial basis function kernel.

Additionally, one noteworthy advantage of the GP model is that it provides predictions with uncertainty information. In other words, the GP outputs both mean predictions and their uncertainty information by providing the standard deviation of the predictions. For a more uncertain predictions, the model returns a higher standard deviation. Conversely, for more confident predictions, the model returns a lower standard deviation. An illustration of the GP model is shown in Fig. 3.1 with fitting results on a noise-free (left) and noisy (right) dataset.

3.1. Hyperparameter optimization

For the hyperparameter optimization of all models, the same random split is used in the training/testing database. This is achieved by fixing the seed (random number) in the code. That allows the models' performances to be evaluated using the same datasets.

3.1.1. SVR

For the SVR model, four hyperparameters were optimized: (1) the kernel, which is the function used in the algorithm to transform (to map) the data into a higher dimension; (2) the C value, which is proportional to the inverse of the regularization parameter and balances the importance of the slack variable errors in the model's optimization; (3) the gamma value, which is a coefficient associated with three kernel functions, radial basis function (rbf), sigmoid, and polynomial; and (4) ε, which specifies the insensitive tube margin thickness.

The four available kernels were evaluated separately: linear, polynomial, sigmoid, and radial basis function (rbf). The C value was optimized for 400 different possibilities: from 0.01 to 1 in increments of 0.01, and from 1 to 300 in increments of 1. The gamma value was optimized for 120 different possibilities: from 0.01 to 1 in increments of 0.01, and from 1 to 20 in increments of 1. ε was optimized for 1000 different possibilities ranging from 0.001 to 1 in increments of 0.001. Note that the polynomial function is a special function since the degree must be specified. For this kernel, all degrees from 2 to 9 were evaluated separately. The grid search tool was used to evaluate all the different combinations. Table 3.1 presents the best-performing hyperparameters for SVR.

3.1.2. RF

For this model, three hyperparameters were optimized: (1) the number of estimators, which determines the number of decision trees (DT) that will be created and trained in the model; (2) the maximum features' number, which determines the number of randomly selected features used in the training of each individual DT; and (3) the maximum samples' number, which determines the size of the database created to train each DT in the model. Number of estimators was optimized for 19 different possibilities: 100 to 1000 in increments of 50. The maximum features' number was optimized for the two options provided: "sqrt" and "log2". The maximum samples' number was optimized for 4 different possibilities, from 0.7 to 1.0. Since there are a total of only 152 different possibilities for optimizing the RF model, all of them were evaluated using the grid search method. Table 3.2 presents the RF hyperparameters with the best performance.

3.1.3. MLP/NN

For this model, four hyperparameters were optimized: (1) the model optimizer, which is responsible for the weights and biases updates after each batch iteration; (2) the number of hidden layers, which determines the size and complexity of the neural network (NN); (3) the number of neurons in each hidden layer, which determines the depth and complexity of the NN; and (4) the neuron activation function in each hidden layer, which is responsible for the non-linearization of the NN.

Three model optimizers were evaluated separately: RMSprop, Adam, and Nadam. They have different methods for updating the NN weights and biases. A randomized search tool was implemented for each of these three evaluations because the search space for a NN can be extremely large and computationally infeasible. To narrow the search space, the number of hidden layers was maintained between 2 and 10, with 20 to 50 neurons per layer in increments of 5 neurons, and three possible activation functions for each layer (sigmoid, relu, and tanh), which gives a total of around 24,000,000 possibilities. Then, a randomized search tool evaluated 3,000 random possibilities for each model solver. Table 3.3 summarizes the results of the three best-performing MLP architectures.

3.1.4. GP

For this model, two hyperparameters were optimized: (1) the kernel function, which is the function used in the algorithm to fit the GP to the data; and (2) α, which is a value used to improve numerical stability while fitting the kernel function. The kernel was optimized for 3 different possibilities: "rbf", "rbf + white", and "matern", while α was optimized for 4 different possibilities: 10^{-10}, 10^{-5}, 0.1, and 0.5. Since there are a total of only 12 different possibilities for the GP model, all of them were evaluated using the grid search tool. Table 3.4 presents the

Table 2.3
Input parameters and their related ranges.

External layers	Dimensional feature	Average Roughness [μm]	Static Contact Angle [°]	Surface Orientation [°]	CHF [kW.m^{-2}]	Reference
Bare Zirlo	7.968	0.067–0.12	64–89	0	571–709	(Jo et al., 2019)
Bare SS304						
Cr-coated Zirlo						
FeCrAl-coated Zirlo						
Bare Zirlo	7.968	0.07–5.72	33–107	0	454–931	(Yeom et al., 2020)
Cr-coated Zirlo						
FeCrAl-coated Zirlo						
SiC						
Bare FeCrAl C26M	1.476–1.484	1.55–25.21	54—82	0	381–922	(He et al., 2022)
Bare FeCrAl B126Y						
Bare FeCrAl B136Y						
Bare Zircaloy-4						
Bare Zirlo						
Bare Zr 705						
Bare Inconel 400						
Bare Inconel 600						
Bare Inconel 625						
Bare SS304						
Bare SS316						
Bare SS321						
Bare SS347						
Bare Titanium-2						
Bare Titanium-9						
Oxidized samples						
Cr-coated sapphire	4	0.01–0.31	0–85	0	980–2220	(Zhang et al., 2023)
Al2O3-coated sapphire						
ITO-coated sapphire						
CuO-coated sapphire						
ZnO-coated sapphire						
SiO2-coated sapphire						
Bare SS304L	4	0.093–0.107	0–81	0	705–1041	(Kim et al., 2019)
Cr-coated samples						
Oxidized samples						
Bare Fe13Cr4Al	3.968	0.32–0.61	68–75	0	784–1089	(Ali et al., 2018)
Bare Zircaloy-4						
Fe13Cr4Al PWR						
Fe13Cr4Al BWR-NWC						
800 grit pol. SS316L	11.916	0.02–0.18	0–68	90	715–947	(Son et al., 2019)
60 grit pol. SS316L						
Fe20Cr5Al-coated samples						
Cr-coated samples						
Zircaloy-4	1.492	0.088	85	0	635	(Seo et al., 2015)
SiC	1.6	0.107	93		1037	
Bare Zircaloy-4	6	0.021–0.028	19–81	0	660–1223	(Kam et al., 2015)
Bare SS						
SiC-coated samples						
Cr-coated samples						
Bare FeCrAl C26M	11.811–11.874	2.12–6.84	64–82	90	608–1062	(He and Chen, 2023)
Bare FeCrAl B126Y						
Bare FeCrAl B136Y						
Bare Zircaloy-4						
Bare Zirlo						
Bare Zr 705						
Oxidized samples						
Bare Zircaloy-4	3.968	0.12–0.66	68–83	0	677–791	(Ali et al., 2020)
Cr-coated samples						
Irradiated samples						
Polished samples						
Bare Zircaloy-4	8	0.154–0.207	0–65	0	804–1004	(Cheol Lee et al., 2019)
ZrSi2-coated Zr-4						
Oxidized samples						
Bare Zircaloy-2	3.051	1.8–4.4	41–59	0	860–1350	(Fong et al., 2003)
Polished samples						
Bare Zircaloy-4	4	0.05–0.32	0–49	0	1004–1924	(Ahn et al., 2010)
Oxidized samples						

GP hyperparameters with the best performance.

4. Results and discussion

After tuning each of the aforementioned ML models, their performances were compared and only the model with the best fitting was selected to provide predictions. Next, feature importance models were used to rank the features in relation to their importance, and then the features' effects on CHF were discussed.

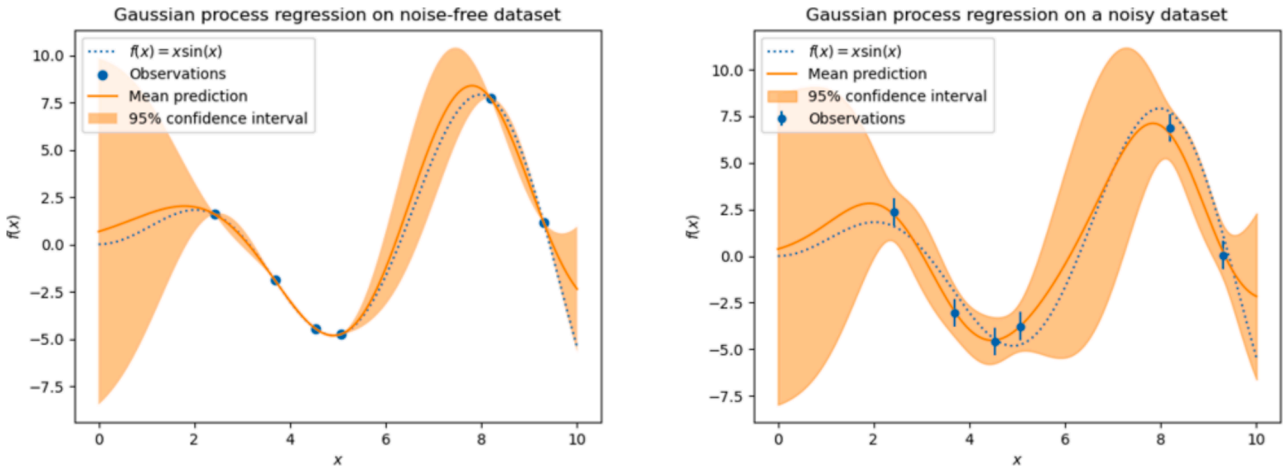


Fig. 3.1. Gaussian Process regression illustration for (a) a noise-free dataset and (b) a noisy dataset (Pedregosa et al., xxxx).

Table 3.1
SVR hyperparameter optimization results.

Kernel	C value	Gamma	ε
rbf	296	0.51	0.411

4.1. Machine learning Models' performance comparison

The models described in the previous section were built and evaluated for 1000 different splits in the database. The database was split using a random split into 80 % training and 20 % testing. The performance of the four ML models is evaluated on three error metrics: the mean absolute error (MAE), mean squared error (MSE), and the mean absolute percentage error (MAPE) in Fig. 4.1. Note that smaller values are desirable for all metrics as they indicate a smaller deviation from the true values.

When comparing the ML models, it can be observed that the RF model had the best performance while the MLP model had the worst performance. MLP also presented the highest variability (smaller precision) with the different database splits, which means the model does not fit well to this database and/or is not generalized. A possible explanation for the MLP's poor performance is that MLP generally requires a large amount of data to leverage its full predictive performance. Another common issue is the MLP can overfit the dataset and have poor performance on the test set. To overcome this, the authors made efforts to relieve overfitting (*e.g.*, early stopping and dropout), but these measures did not significantly improve performance. On the other hand, all other models show very small variabilities (higher precision) in their performances, meaning these models are more generalized. Since the RF model performed best, it was selected to be the focus of the following analyses.

4.2. Feature importance models

In order to rank the effect of each input parameter on predicting the CHF, four different feature importance models were built and applied to the RF model. Before performing those evaluations, an initial purely qualitative analysis was performed by generating a correlation matrix among the different features, as well as their effects on CHF, Fig. 4.2.

Table 3.2
RF hyperparameter optimization results.

Number of trees	Number of features	Maximum samples
200	log2	0.9

Table 3.3
MLP hyperparameter optimization results.

Model Solver	Number of NN constructed	Number of hidden layers	MLP structure	MLP activation functions
RMSprop	3,000	2	6–45-30–1	relu-tanh-relu
Adam	3,000	2	6–50-40–1	relu-tanh-relu
Nadam	3,000	7	6–20-45–30–45–35-30–45-1	relu-tanh −tanh −tanh −relu-tanh-sig-sig

First, a column of random numbers was inserted into the database to serve as a baseline. This random feature is assumed to have no effect on CHF prediction. If a feature behaves similarly to the baseline, then it has a minimal effect on CHF prediction. On the other hand, if a feature does not behave like the baseline, then it has a more crucial effect on CHF prediction.

The red circles highlight the important outcomes of the correlation matrix. First, the random number feature, as expected, has a near zero effect on CHF. However, what was not expected was that surface orientation presented no effect on CHF. That can be explained by the fact that 89.11 % of the datapoints considered a horizontal upward facing heating element (0°) while the other 10.89 % considered a vertical heating element (90°), which makes this feature very sparse. Last, a very strong correlation is shown between the surface orientation and the dimensional feature, which was expected because different considerations were taken for vertical and horizontal samples.

4.2.1. Mean decrease impurity (MDI) model

Also known as Gini importance, this model is an intrinsic feature importance model available in the created random forest regressor model. It sums all feature importance across the nodes of the trees and then averages them for all trees, Fig. 4.3.

As seen in Fig. 4.3, it follows the initial results of the correlation matrix, where surface orientation has no effect on CHF, since its importance is lower than the random number. An important outcome from this model is that pressure and static contact angle together account for almost 50 % of the effect on CHF. These quantities will be confirmed with the results of the other three models.

Table 3.4
GP hyperparameter optimization results.

Kernel	α
matern	0.5

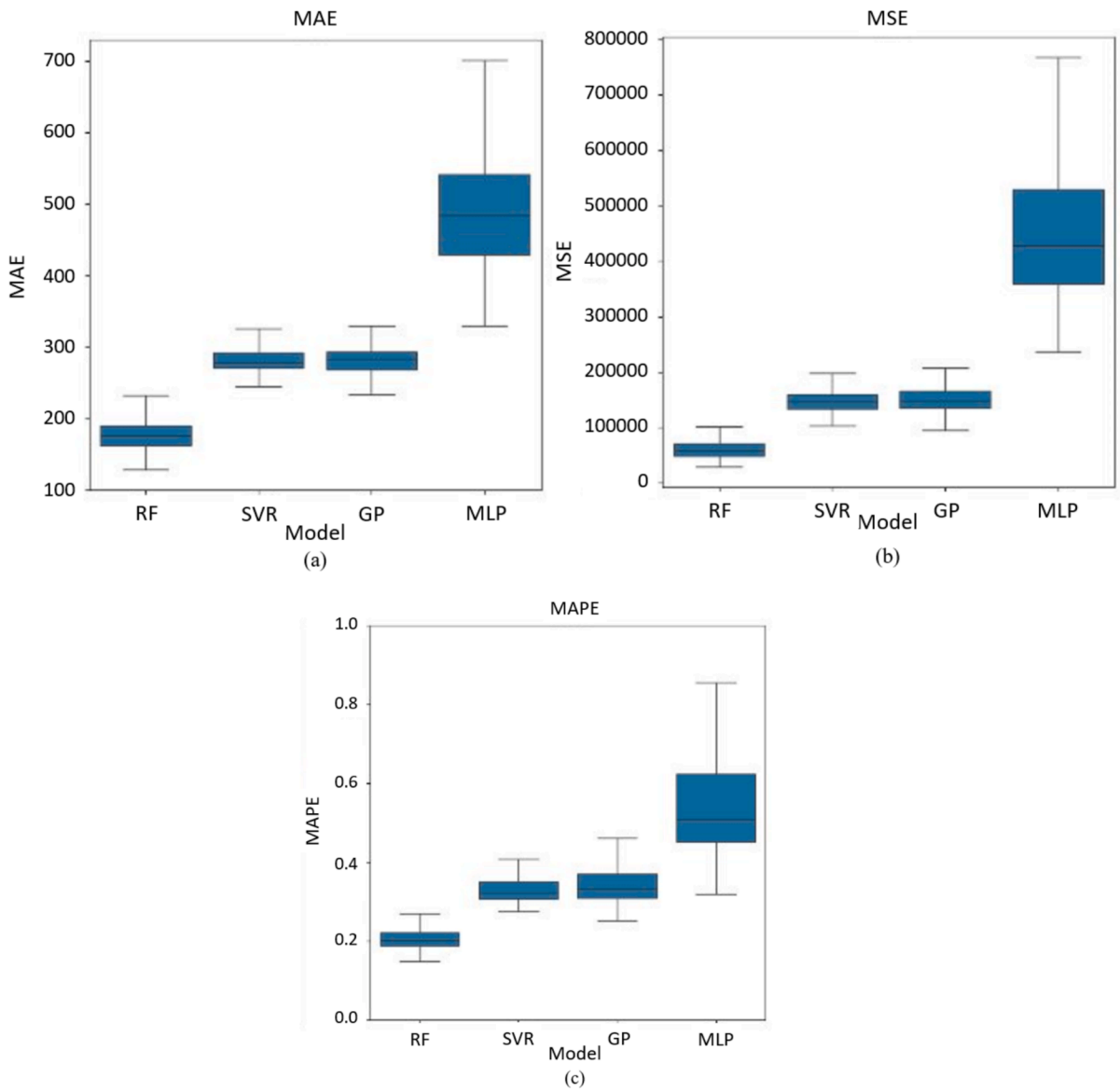


Fig. 4.1. ML models' performance comparison using (a) MAE, (b) MSE, and (c) MAPE.

4.2.2. Permutation model

The permutation model works by shuffling one feature at a time, randomly rearranging an entire column with the existing data, and checking how much each feature impacts the fitting loss of the model, Fig. 4.4.

Permutation results corroborate with the correlation matrix and the MDI results. For this model, the combined effect of pressure and static contact angle on CHF was even higher than 50 %.

4.2.3. Drop column model

A baseline loss with all the features is first calculated. Then, the loss is recalculated by dropping out each of the features. Ideally, a significant feature will have the largest decrease/change in the loss, while an insignificant feature will have a small change in the loss, Fig. 4.5.

Drop column results are slightly different from the previous models, as the dimensional feature surpasses the pressure importance. However, it still ranks surface orientation as the least important feature.

4.2.4. SHAP values

SHAP values can be used to interpret the importance of a feature on the ML model outputs. SHAP values come from Shapley values from game theory (Hart, 1989). The impact of each feature is compared against the average, Fig. 4.6.

SHAP results confirm what has been observed for previous importance models: static contact angle and pressure are the most important features, while surface orientation has no effect on CHF for this dataset. In relation to static contact angle, the number of red datapoints on the negative side of SHAP values means this parameter has a negative effect on CHF, *i.e.*, increasing static contact angle decreases CHF. Based on a similar analysis, it is easy to observe that pressure has a positive effect on CHF, since the majority of red dots sit on the positive SHAP values. Among the other five features, surface thermal effusivity is the only one interpretable, suggesting a negative effect on CHF. The effect of the other features cannot be determined by their SHAP values as possible combined effects might be important.

Therefore, the discussion of the parametric effects on CHF in the next section is essential to explain and support the feature importance models' results obtained in this section.

4.3. Parametric effects on CHF

In this section, a deep discussion of the effect of each input parameter

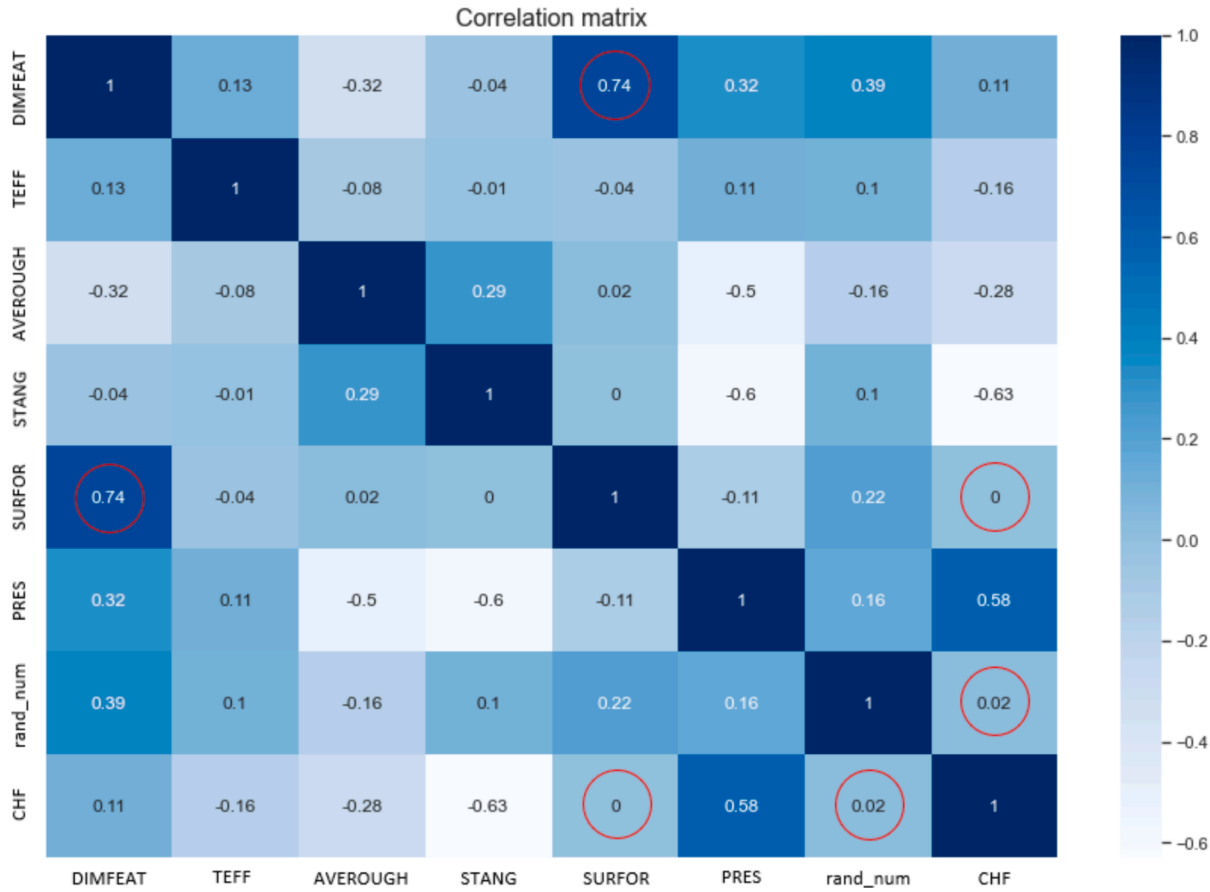


Fig. 4.2. Database correlation matrix.

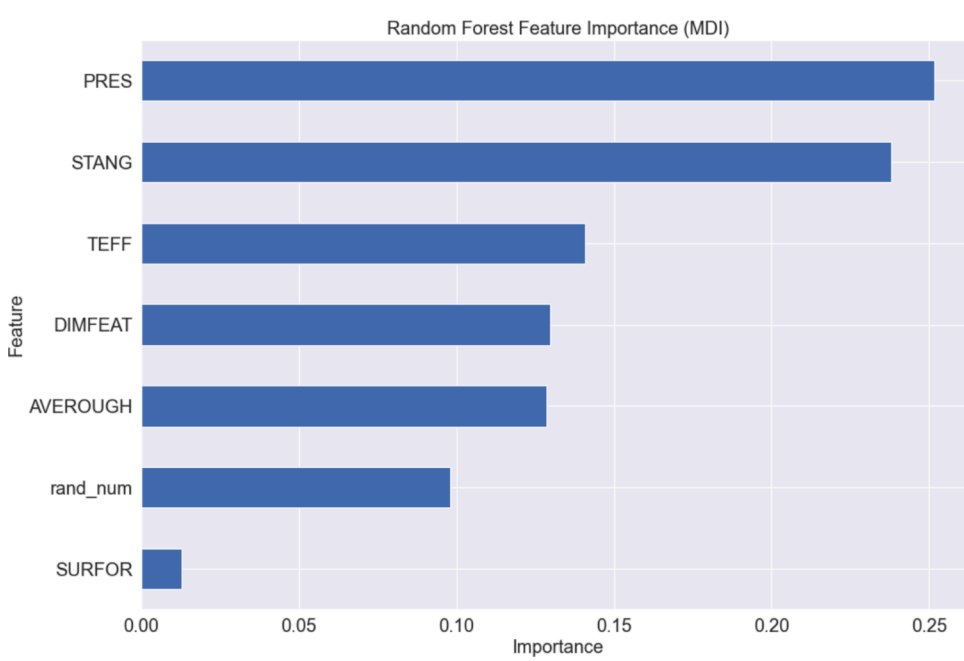


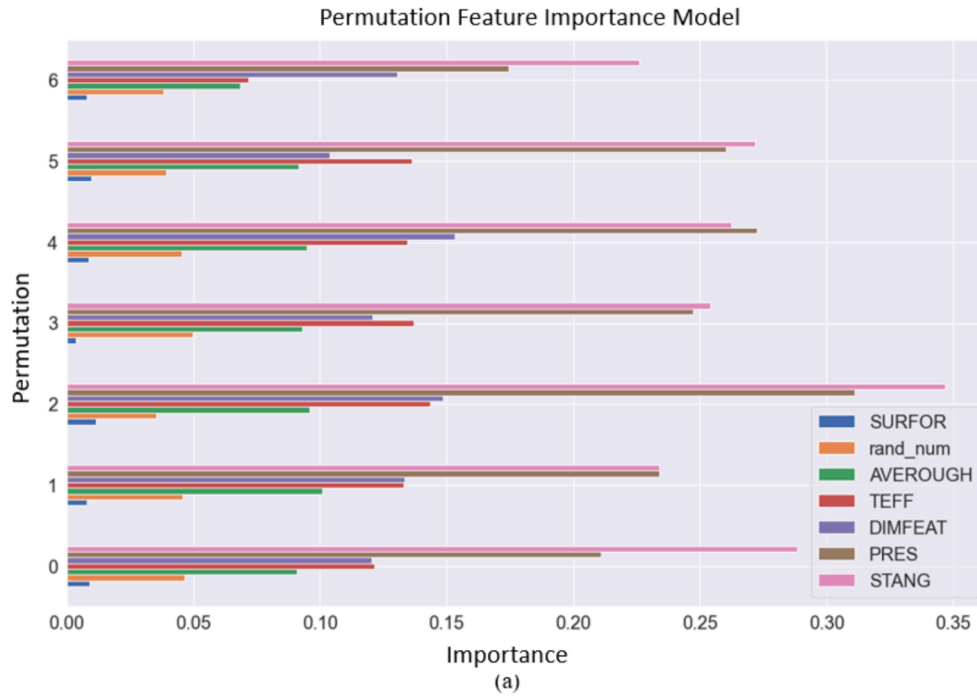
Fig. 4.3. MDI feature importance model result.

on pool boiling CHF was taken.

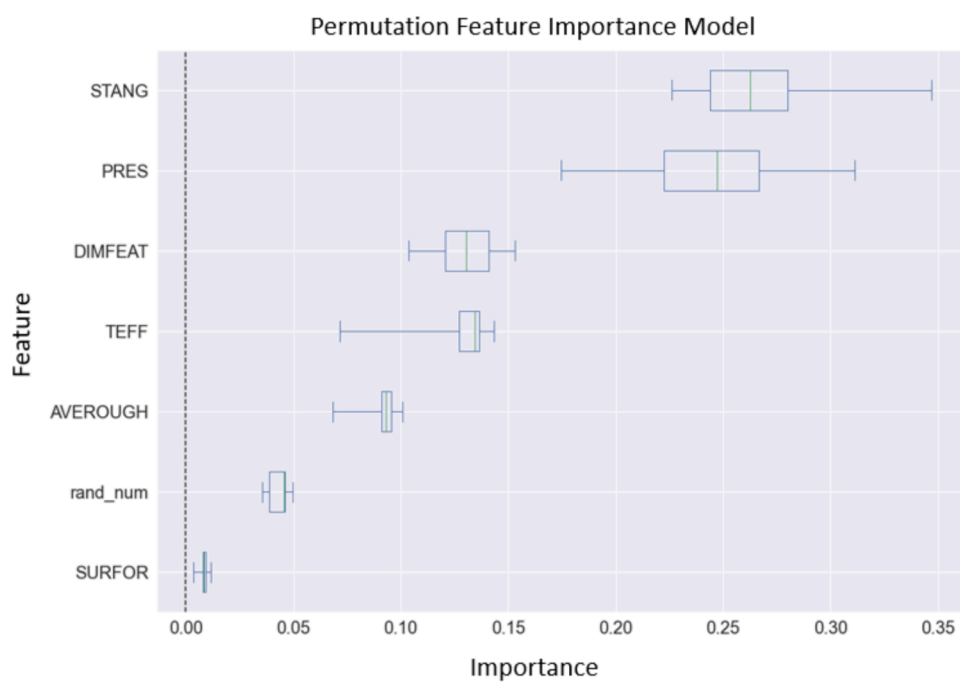
4.3.1. *RF model construction and predictions*

The RF model performance after its implementation procedure, *i.e.*, decision trees training and their subsequent testing, is shown in Fig. 4.7.

The performance shown in Fig. 4.7 is summarized in Table 4.1. The model achieved a 95 % R^2 score for the training dataset and an 89 % R^2 score for the testing dataset. The ratio between predicted and



(a)



(b)

Fig. 4.4. Permutation feature importance model result for (a) each permutation, and (b) compiled.

experimental values for the testing dataset remains in ± 20 % error range, Fig. 4.7 (b). The RF model's CHF predictions were plotted with experimental CHF for the different static contact angles and surface orientations. Kandlikar's model predictions were also included in the plot, Fig. 4.8.

It is very important to emphasize that Kandlikar's model considers the receding contact angle for its predictions, while static contact angles were collected from the literature to build the ML database. Hence, the presence of both Kandlikar's predictive curves in Fig. 4.8 is to show the CHF trend in relation to contact angle but cannot be used to compare the RF predictions to the experimental datapoints. Despite this, it can be observed that RF predictions are very close to the experimental data. Furthermore, when the static contact angle is 0 °, experimental CHF ranges from 813 to 2220 kW.m^2, which suggests a new method, other than the widely used liquid droplet contact angle, should be considered to investigate the material's wettability when testing extremely hydrophilic materials. In these cases, surface tension forces are enhanced, which increases the surface's rewetting speed. Therefore, rewetting speed measurement methods have been applied (Son et al., 2020; Son et al., 2018; Son and Kim, 2019).

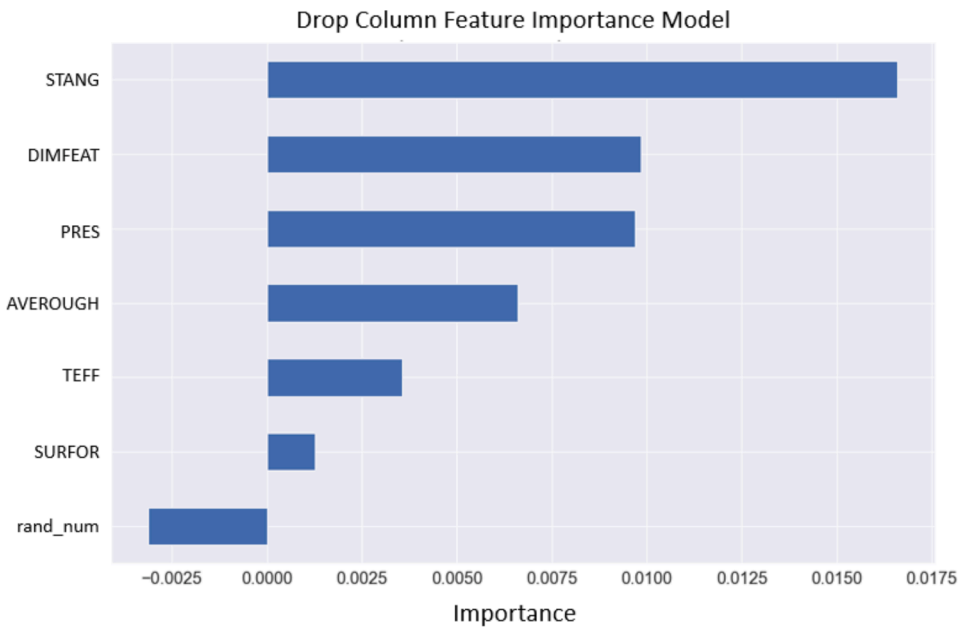


Fig. 4.5. Drop column feature importance model result.

Fig. 4.6. SHAP values.

4.3.2. Pressure vs CHF

The effect of pressure on pool boiling CHF has been well-understood for a long time. Considering Zuber (Zuber, 1959); Kutateladze (Kutateladze, 1952), and Rohsenow (Rohsenow, 192AD) correlations, CHF increases until a maximum value, around 7 MPa, and then decreases until the saturated vapor pressure, Fig. 4.9.

This behavior is explained by the competition between the fluid properties considered in the correlation, at different pressures, *i.e.*, liquid and gas densities, enthalpies, and surface tension. In low-pressure ranges, vapor density increases considerably with pressure increasing, while in high-pressure ranges, surface tension and enthalpy of vaporization are the most affected, both decreasing with pressure increase. This leads to a maximum CHF value at the point where the increase in CHF from the increasing vapor density is off-set by the decrease in CHF caused by the decreasing surface tension and enthalpy of vaporization.

In relation to the effect of pressure on the pool boiling curve, an increase in the slope is observed with pressure, meaning as the heat transfer coefficient increases, the curves shift to the left (Dahariya and Betz, 2019). It is also observed that the CHF superheat decreases when pressure increases. This behavior is driven by bubble dynamic changes with pressure. When pressure increases, the bubble departure size decreases due to the change in enthalpy of the surrounding liquid as well as the bubble's specific volume changes. Additionally, bubble departure frequency increases, which is explained by the surface tension decreasing with pressure increase, hence the buoyancy force overcomes surface tension force more rapidly, which incurs faster bubble detachments. Furthermore, the number of nucleation sites is also affected positively by pressure, increasing detachment speed, due to more bubbles coalescing, hence helping to delay CHF.

The RF model was able to capture the pressure effect on CHF, Fig. 4.10, even though the pressure data range is small, 83,8 kPa to 101,2 kPa, which represents a change of around 100 $kW.m^{-2}$ (6–10 %) using Zuber CHF correlation.

The fact that pressure was selected as one of the most important features can be explained by the database normalization step which has potentialized this effect.

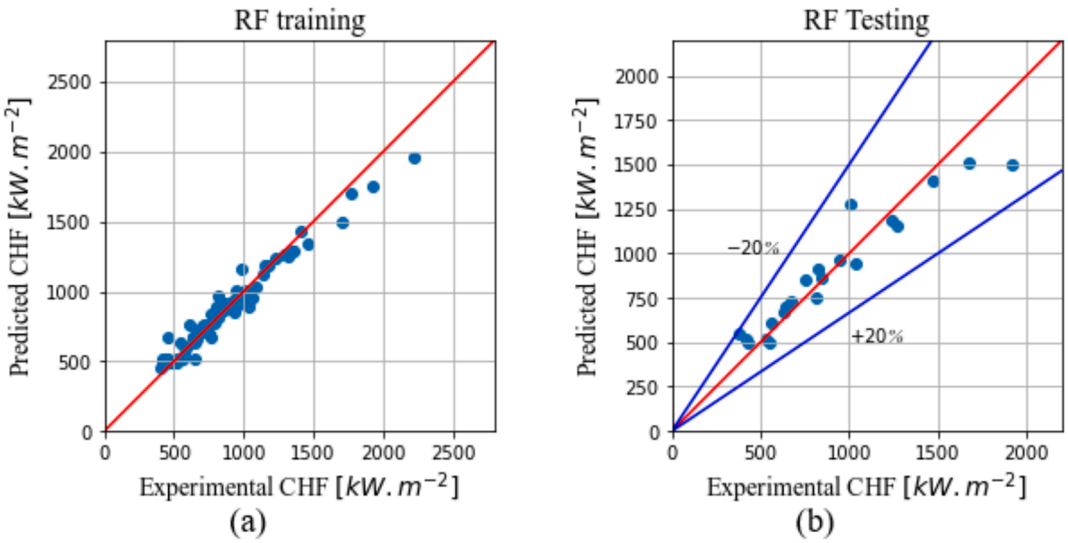


Fig. 4.7. RF model performance (a) training dataset, and (b) testing dataset.

Table 4.1
RF model scores summary.

	Training dataset	Testing dataset
R^2	0.952	0.893
MAE	0.149	0.267
MSE	0.044	0.134
MAPE	0.559	0.511

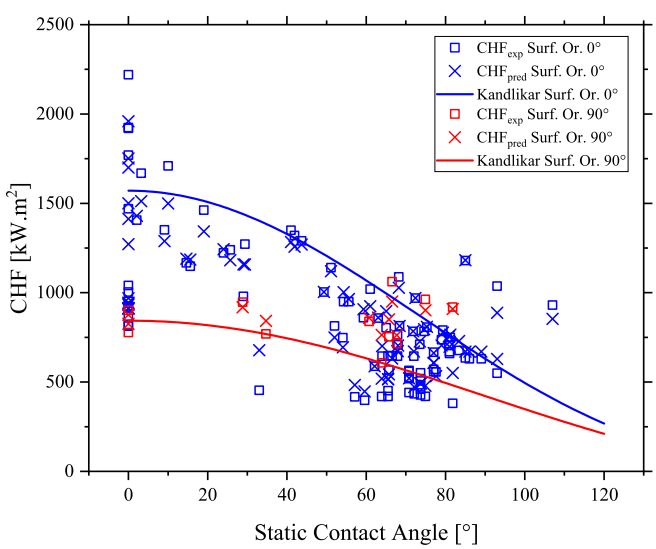


Fig. 4.8. Experimental vs Predicted CHF data.

4.3.3. Static contact angle vs CHF

In the past 50 years, it has been observed and it is well accepted that pool boiling CHF is inversely proportional to the surface contact angle, also known as wettability. Kandlikar's model given by Eq. 1, implements surface contact angle on CHF predictions and has been the basis for many authors' comparison in the past 20 years. However, it shows low data agreement for super hydrophilic surfaces suggesting some mechanisms were not considered in its assumptions.

Ahn *et al.* (Ahn et al., 2011) suggested for super-hydrophilic surfaces (contact angle smaller than 10°) surface rewetting happens much faster than for hydrophilic surfaces, due to a suction force provided by micro, nano, and/or micro-nano structured surfaces. Based on this assumption,

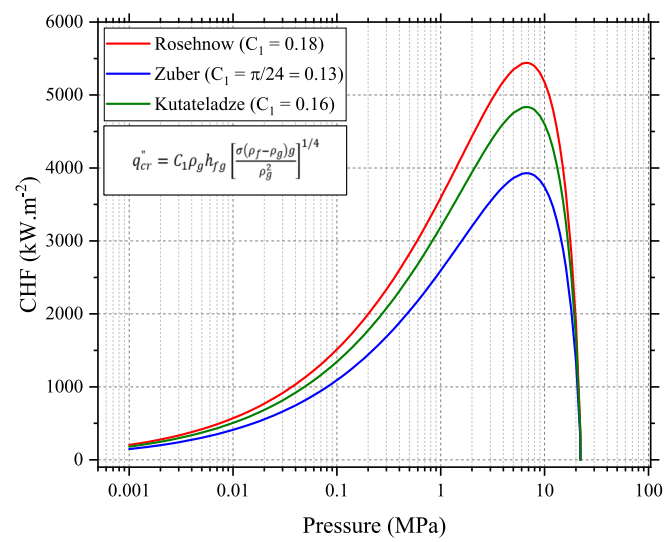


Fig. 4.9. Pressure vs CHF for water-based pool boiling experiments () adapted from Todreas and Kazimi, 1993.

an improved CHF predictive model was developed where a dynamic droplet-motion analysis was considered to determine the amount of absorbed liquid into the outermost surface layer, in other words, a liquid spreading contribution was added to Kandlikar's predictive model. One year later, Ahn *et al.* published a new study where they consolidated their findings from the previous year (Ahn et al., 2012), where the liquid suction was related to the capillary wicking, and the outermost layer thickness (δ) and porosity (ε) were introduced to the new predictive correlation, Eq. 4.

$$q''_{CHF} = S \times h_{lg}\rho_g^{1/2}\left(\frac{1+\cos\beta}{16}\right)\left[\frac{2}{\pi}+\frac{\pi}{4}(1+\cos\beta)\cos\phi\right]^{1/2}\left[\sigma(\rho_l-\rho_g)g\right]^{1/4} + \frac{\varepsilon\delta\rho_l h_{lg}K}{A_{heating}}\frac{dA_{wetted}}{dt}$$

where S is a heating element area correction factor, $A_{heating}$ is the area of the heating surface, A_{wetted} is the area wetted by capillary wicking, dA_{wetted}/dt is the wetted area time derivative, defined as capillary wicking action, and K is a fitting factor related to the capillary wicking action, equal to 0.55.

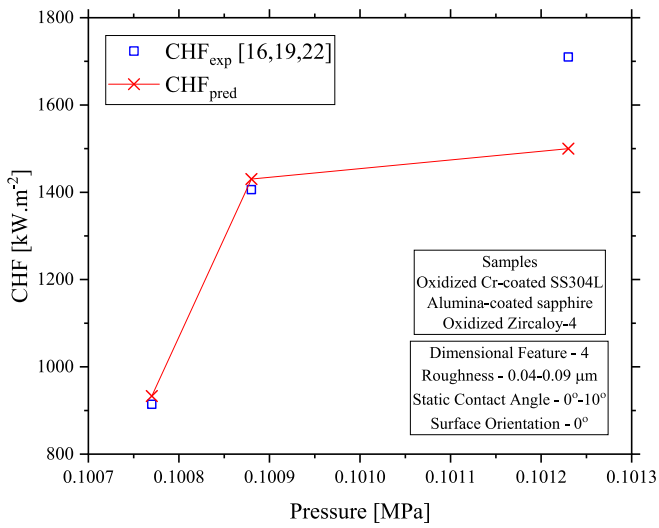


Fig. 4.10. Pressure vs CHF RF predictions.

Two years later, taking a similar approach as Kandlikar, *i.e.*, a force balance on a bubble, and starting from Quéré assumption (Quéré, 2008); Quan *et al.* (Quan et al., 2014) developed a new correlation which corrected Kandlikar's model by adding a wavelength contribution and introducing a critical contact angle, which is calculated based on the roughness ratio (r) and the solid fraction (ϕ_s), which determines a threshold value for the capillary wicking contribution to CHF, Eq. 5.

$$q''_{CHF} = h_{lg}\rho_g^{1/2}\left[\sigma(\rho_l-\rho_g)g\right]^{1/4}\left(\frac{1+\cos\beta}{16}\right)\left[\frac{2}{\pi}\left(1-\sqrt{\phi_s}\right)^{-\frac{1}{2}}\left(\frac{r+\cos\beta}{1+\cos\beta}\right)+\frac{\pi}{4}\left(1-\sqrt{\phi_s}\right)^{1/2}(1+\cos\beta)\cos\phi\right]^{1/2}$$

The inverse relation between CHF and contact angle showed by the RF model predictions, Fig. 4.11, is consistent with all previous findings and discussions.

4.3.4. Average roughness vs CHF

Jones *et al.* (Jones et al., 2009) presented a state-of-the-art study discussing the roughness effect on nucleate pool boiling heat transfer. The inefficacy in correlating roughness parameters, such as average roughness (R_a), rms roughness (R_q), maximum profile peak height (R_p), and five-point average maximum height (R_z), with nucleate boiling heat transfer coefficient (h) behavior was addressed, pointing out that a relation of the form $h \propto R_x^m$ had been previously observed. In order to clarify the importance of roughness to nucleate boiling heat transfer, a series of experiments was performed using two fluids, water and FC-77, and surfaces with different roughness, ranging from 0.027μ m to 10.0μ m, obtained by two different processes: sandpaper polishing and electrical discharge machining (EDM). They compared their experimental results with predictions from three correlations (Cooper, 1984; Cooper, 1984; Gorenflo, 1993; Leiner, 1994) showing those models provide good predictions depending on both the fluid used as well as the roughness of the surface.

Despite the great work, the aforementioned investigation solely focused on roughness, failing to consider the strong correlation between contact angle and roughness. In fact, the potential effect of contact angle in the m factor is addressed, but not explored. Taking a different approach, Kim *et al.* (Kim et al., 2018) also investigated the effect of roughness on pool boiling heat transfer but focused on hydrophobic surfaces. This time, the roughness-contact angle relation is explored, which is emphasized by the concepts of hydrophobic, hydrophilic, and super-hydrophilic surfaces. Based on the experimental results, they proposed a CHF correlation for hydrophobic surfaces, which has an exponential behavior based on the cosine of the surface contact angle.

Taking the above discussion into consideration, the roughness effect on pool boiling CHF cannot be easily determined. Even though Jones *et al.* observed a positive relation between roughness and CHF, other authors observed a maximum boiling heat transfer improvement with roughness (Vachon et al., 1968), while the present authors believe a combined roughness-contact angle factor should be investigated as a

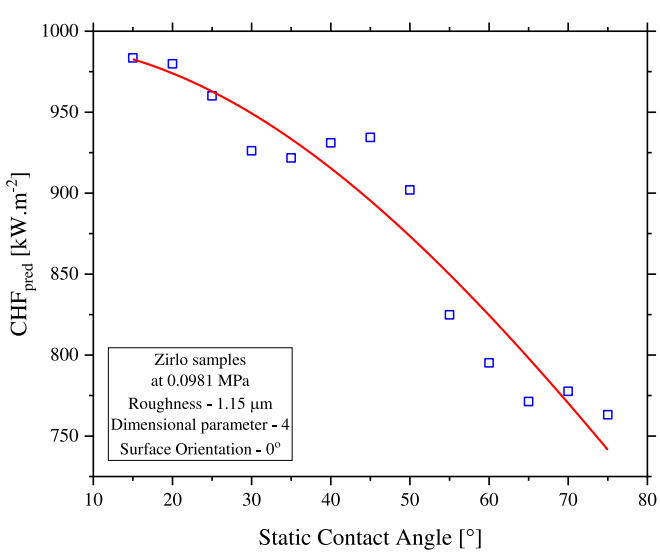


Fig. 4.11. Static Contact Angle vs CHF RF predictions.

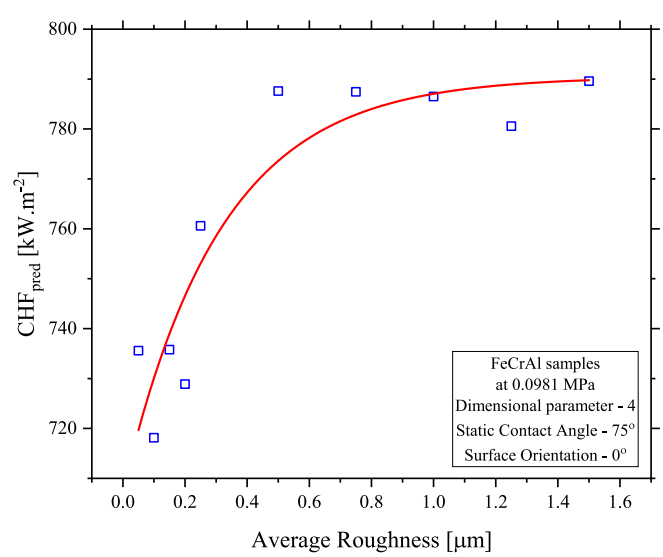


Fig. 4.12. Average Roughness vs CHF RF predictions.

factor to improve CHF correlations, as also suggested by other studies where roughness is related to capillary wicking (Ahn et al., 2011; Quan et al., 2014).

RF predictions based on different average roughness is shown in Fig. 4.12. It follows the understanding of an existing maximum heat transfer improvement, however, as discussed, this is not common sense.

4.3.5. Dimensional feature vs CHF

The effect of different sample sizes on CHF was investigated by a few studies (Pattanayak et al., 2021; Yang et al., 1997; Winter and Weibel, 2023). This effect is more predominant when the surface is facing downwards (as the vapor bubbles and columns get trapped on flat surfaces) compared to when it is facing upwards, when those same vapor bubbles and columns can freely move away from the surface, allowing surface rewetting. These previous studies show that CHF decreases when increasing the heating element's dimensions (Pattanayak et al., 2021; Yang et al., 1997), suggesting the phenomena can be better explained by the macrolayer dryout model (Haramura and Katto, 1983) instead of the hydrodynamic instability model proposed by Zuber (Zuber, 1959).

In a study focusing on investigating the effects of fin height and array spacing on heat transfer performance (Winter and Weibel, 2023), authors hypothesize normalized dimensional parameters using the capillary length (Eq. 2) as the normalizing factor. This is quite an interesting and clever assumption since the capillary length represents the surface tension and the body (gravity) forces balance acting on the interface. Prior to this study, Rahman *et al.* (Rahman et al., 2015) investigated how creating strip divisions in the heating element would affect heat transfer. They observed that decreasing the pitch size between those divisions enhanced heat transfer performance until an optimal value: a pitch size very close to the capillary length. They concluded that if the pitch is equal to the capillary length, the bi-conductive surface achieves the best heat transfer performance. Voglar *et al.* (Voglar et al., 2018) used this work as a base to investigate other possible arrangements. They modified some samples to increase the number of active nucleation sites, obtaining the best performance for a sample where the modifications were two-dimensional 2.5 mm spaced circles with 2 mm in diameter, achieving a CHF improvement factor of 3.7 in relation to a bare surface CHF.

The RF model predictions provided by keeping all other features constant while varying the dimensional feature, Fig. 4.13, shows an unclear trend. The different samples' geometries collected in this study might be the reason why a clear trend was not observed. The lowest dimensional feature values, smaller than 2, refer to wire samples, while the highest ones, greater than 10, refer to tubes, for example. Those effects were not captured by any other parameter.

4.3.6. Thermal effusivity vs CHF

In 1967, Tachibana (Tachibana et al., 1967) performed a few hundred pool boiling experiments to study the non-hydrodynamic effects on CHF. Flat vertically oriented samples made of different materials with different thicknesses were tested. By analyzing a new parameter, heat capacity per unit area, it was suggested that there is a thickness threshold value where heat capacity does not have any significant effect on the sample anymore. CHF reduces with decreasing this parameter, while CHF achieves an asymptotic value with increasing this parameter.

Almost 40 years later, Arik and Bar-Cohen (Arik and Bar-Cohen, 2003) confirmed what was observed by Tachibana while investigating another parameter, thermal activity (the thermal effusivity multiplied by the material thickness). Since thermal effusivity translates the intensity of energy transfer on the interface between two mediums, it is highly affected by a material's properties and temperature.

Recently, He *et al.* (He et al., 2022) showed the thermal effusivity ratio of different materials with a specific fluid does not present a clear trend. By introducing a new parameter, nucleate boiling surface stability (NBSS), and showing it has a linear trend with CHF, the effects of thermal effusivity ratio, thermal diffusivity ratio, and the receding contact angle were coupled together, shown in Eq. 6.

$$NBSS = \left(\frac{\sqrt{(k\rho c_p)_{wall}}}{\sqrt{(k\rho c_p)_{fluid}}} + 1 \right) \frac{\left(\frac{k}{\rho c_p}\right)_{fluid}}{\left(\frac{k}{\rho c_p}\right)_{wall}} (1+\cos\theta_r)\left(\frac{2}{\pi}+\frac{\pi}{4}(1+\cos\theta_r)\right)^{0.5}$$

In agreement with He *et al.* results, Fig. 4.14. shows the RF predictions do not provide a clear trend for larger thermal effusivities, but it seems to present a decreasing exponential behavior.

4.3.7. Surface orientation vs CHF

As introduced in section 4.3.5, increasing surface orientation from facing upwards (0°) to facing downwards (180°), passing by vertically oriented (90°), decreases pool boiling CHF due to vapor getting trapped, which prevents surface rewetting. Among a few tens of studies that have investigated the effects of surface orientation on CHF, just a few will be specially addressed due to their major contribution to the scientific community.

In 1965, Lyon (Lyon, 1965) performed a series of saturated pool boiling experiments using a platinum disk as the heating element, with

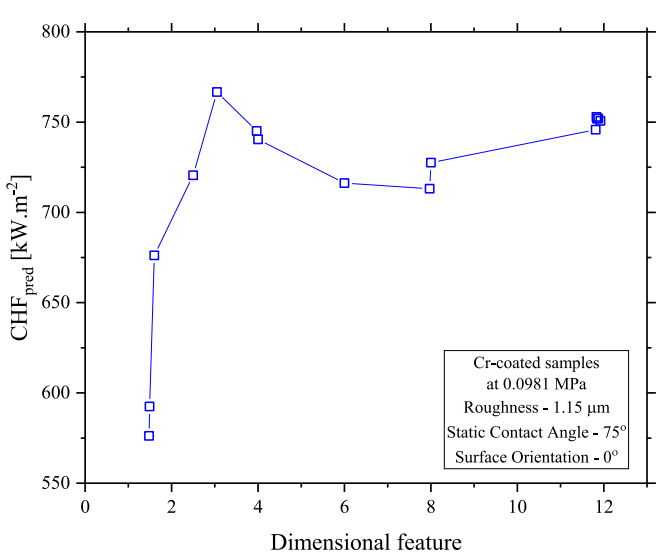


Fig. 4.13. Dimensional feature vs CHF RF predictions.

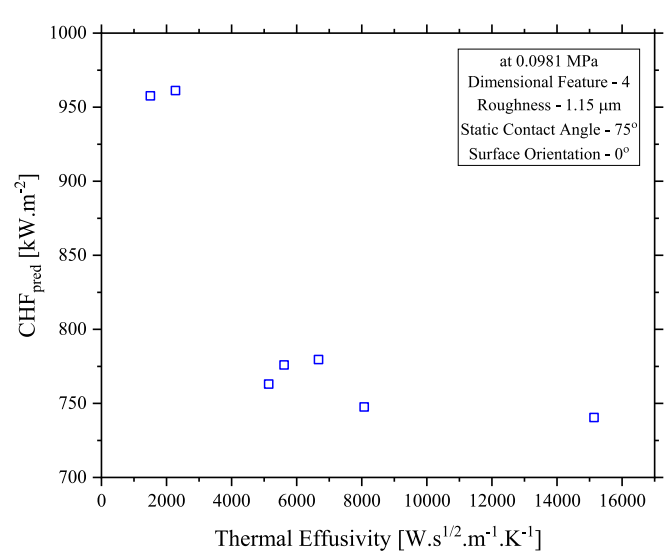


Fig. 4.14. Thermal effusivity vs CHF RF predictions.

pressures ranging from 6.0 to 228.0 kPa and with 8 different surface orientations ranging from 0° to 180°. In 1974, a correlation was built using Lyon's experimental data (Vishnev, 1974). Another correlation was built in 1993 (El-Genk and Guo, 1993), which was developed for three different liquids: water, nitrogen, and helium. This correlation was derived from the CHF prediction model developed by Kutateladze (Kutateladze, 1952), which was later improved by Zuber.

By performing an imaging investigation of the vapor behavior at different orientations, Howard and Mudawar (Howard and Mudawar, 1999) observed there are three different boiling mechanisms. They suggested dividing the surface orientation into three different regions, which relate to the different boiling mechanisms: (1) upward-facing region (0°-60°), where bubbles vertically detach from the surface allowing the surface to rewet, (2) a near-vertical region (60°-165°), where bubbles move along the surface before detaching presenting a wavy behavior due to Kevin-Helmholtz instabilities, and (3) downward-facing region (165°-180°), where vapor stays trapped under the surface, thus rewetting is impossible. This theory suggests that a single CHF predictive model is not possible due to the three different regions presenting three different boiling mechanisms.

Two years later, Kandlikar (Kandlikar, 2001) developed a saturated pool boiling CHF correlation which can be applied to surfaces with orientations from 0° to 90° (Eq. 1). This emphasizes that heating element shape and size affect CHF, particularly for near vertical and facing downwards, which were not considered by Howard and Mudawar since they tested heating elements with different shapes and sizes.

Because the feature importance models' results showed surface orientation does not have an effect on CHF, the RF model is not able to provide predictions for different surface orientations. This is due to the fact that the datapoints contained only 0° and 90° surface orientations. If more data in between these two orientations are collected to improve the database in the future, the model could be trained again, allowing it to provide predictions for different surface orientations.

After evaluating the RF model by analyzing each parametric effect individually, the model showed to be able to capture most of the features' trends. This emphasizes the accuracy of this predictive model since the available correlations fail to provide accurate predictions when surface characteristics are involved.

5. Conclusion

This study aimed to use ML models as a tool to quantitatively rank the parametric surface effects on CHF of ATF materials in pool boiling experiments. To reach this goal, the following steps were taken:

- A pool boiling CHF database was built by gathering 101 experimental datapoints from 14 studies focusing on ATF materials available in the literature.
- Selection of pressure, dimensional feature, average roughness, static contact angle, surface orientation, and heating element thermal effusivity as input parameters (features) to be used in the ML models training.
- Database preprocessing (datapoints inspection, database normalization, database split in 80 % training and 20 % testing).
- Selection and implementation of SVR, RF, SVR, and GP ML models.
- ML models' hyperparameter optimization (tuning).
- ML models' performance comparison and analysis.
- Selection of RF as the ML model with the best performance for this database.
- Implementation of feature importance models to the RF model, providing feature ranking.
- Discussion of each feature's effect on CHF.
- RF performance analyses and predictions.

Four ML models were created, trained, and compared: SVR, MLP, RF and MLP. RF presented the best-fitting performance, by showing the smallest errors. In addition, RF also showed to be the most accurate (generalized model), while MLP showed both the worst fitting performance and the worst precision.

The feature importance models agreed that pressure and static contact angle have the highest effect on CHF accounting for about 50 % of the total effects, while surface orientation has no significant effect on CHF for the constructed database.

After the features were ranked, a final RF model analysis was performed. The model showed good fitting values to the database for both training and testing datasets, while its predictions were compared to the experimental data showing the model provide good predictions to ATF pool boiling CHF. Features' behaviors were also evaluated, and the RF model showed to be able to successfully capture their trends.

This shows ML models can, if properly trained and if a minimum performance is achieved, be used as predictive models where there is not a mathematical model, or the available mathematical model involves many intrinsic uncertainties. Furthermore, the performance and analyzes can be drastically improved if more datapoints were available in the literature and other parameters had been published, such as receding contact angle, roughness ratio, *etc*.

CRediT authorship contribution statement

Bruno P. Serrao: Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation. **Ye Kwon Huh:** Writing – original draft, Software, Formal analysis. **Eliot Ciuperca:** Writing – review & editing, Software, Data curation. **Elvan Sahin:** Writing – review & editing. **Kaibo Liu:** Writing – review & editing. **Juliana P. Duarte:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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